CP2.3 — Methods in Calculus

1. Improper Integrals

An integral is **improper** if either:

- One or both limits are infinite: $\int_1^\infty f(x)\,dx$
- The integrand has a discontinuity (vertical asymptote) within the interval: $\int_0^1 \frac{1}{\sqrt{x}}\,dx$

We evaluate by replacing the problematic limit with a parameter and taking a limit.

1.1 Infinite Limits

$$\int_a^\infty f(x)\,dx = \lim_{t\to\infty} \int_a^t f(x)\,dx$$

If this limit exists and is finite, the integral **converges**. Otherwise it **diverges**.

Example: Determine whether $\int_1^\infty \frac{1}{x^2}\,dx$ converges.

$$\int_1^t \frac{1}{x^2}\,dx = \left[-\frac{1}{x}\right]_1^t = -\frac{1}{t} + 1 \xrightarrow{t\to\infty} 1$$

Converges to 1.

Example: Determine whether $\int_1^\infty \frac{1}{x}\,dx$ converges.

$$\int_1^t \frac{1}{x}\,dx = [\ln x]_1^t = \ln t \xrightarrow{t\to\infty} \infty$$

Diverges.

1.2 The p-Test (Critical Result)

$$\int_1^\infty \frac{1}{x^p}\,dx \text{ converges if and only if } p > 1$$

$$\int_0^1 \frac{1}{x^p}\,dx \text{ converges if and only if } p < 1$$

Proof for $\int_1^\infty x^{-p}\,dx$:

For $p \neq 1$: $\int_1^t x^{-p}\,dx = \left[\frac{x^{-p+1}}{1-p}\right]_1^t = \frac{t^{1-p}-1}{1-p}$

- If $p > 1$: $1-p < 0$, so $t^{1-p} \to 0$ as $t \to \infty$. Converges to $\frac{1}{p-1}$.
- If $p < 1$: $1-p > 0$, so $t^{1-p} \to \infty$. Diverges.
- If $p = 1$: gives $\ln t \to \infty$. Diverges.

1.3 Discontinuities within Interval

$$\int_a^b f(x)\,dx = \lim_{\varepsilon\to 0^+} \int_a^{b-\varepsilon} f(x)\,dx \quad \text{if } f \text{ has singularity at } b$$

Example: Evaluate $\int_0^1 \frac{1}{\sqrt{x}}\,dx$.

$$\int_{\varepsilon}^{1} x^{-1/2}\,dx = [2\sqrt{x}]_{\varepsilon}^{1} = 2 - 2\sqrt{\varepsilon} \xrightarrow{\varepsilon \to 0^+} 2$$

Converges to 2.

Exam tip: Always write the integral with a limit parameter, evaluate the definite integral, then explicitly take the limit. Don't just "put in ∞" — show the limiting process.

2. Mean Value of a Function

The **mean value** of $f(x)$ on the interval $[a, b]$ is:

$$\bar{f} = \frac{1}{b-a}\int_{a}^{b} f(x)\,dx$$

Geometrically, this is the height of a rectangle of width $(b-a)$ having the same area as the area under $f(x)$ from a to b.

Example: Find the mean value of $f(x) = x^2 + 1$ on $[1, 4]$.

$$\bar{f} = \frac{1}{4-1}\int_{1}^{4}(x^2+1)\,dx = \frac{1}{3}\left[\frac{x^3}{3} + x\right]_{1}^{4}$$

$$= \frac{1}{3}\left[\left(\frac{64}{3} + 4\right) - \left(\frac{1}{3} + 1\right)\right] = \frac{1}{3}\cdot\frac{63}{3+1}\cdot\frac{1}{1}$$

$$= \frac{1}{3}\left(\frac{64}{3} + 4 - \frac{1}{3} - 1\right) = \frac{1}{3}\cdot(21+3) = \frac{1}{3}\cdot 24 = 8$$

Wait — let me recalculate: $\frac{64}{3} + 4 - \frac{1}{3} - 1 = \frac{63}{3} + 3 = 21 + 3 = 24$. So $\bar{f} = \frac{24}{3} = 8$.

Example: Find the mean value of $f(x) = \sin x$ on $[0, \pi]$.

$$\bar{f} = \frac{1}{\pi}\int_0^{\pi} \sin x\, dx = \frac{1}{\pi}[-\cos x]_0^{\pi} = \frac{1}{\pi}(1+1) = \frac{2}{\pi}$$

3. Differentiation of Inverse Trigonometric Functions

$$\frac{d}{dx}(\arcsin x) = \frac{1}{\sqrt{1-x^2}}, \quad |x| < 1$$

$$\frac{d}{dx}(\arccos x) = \frac{-1}{\sqrt{1-x^2}}, \quad |x| < 1$$

$$\frac{d}{dx}(\arctan x) = \frac{1}{1+x^2}$$

3.1 Proof for arcsin

Let $y = \arcsin x$, so $\sin y = x$. Differentiating implicitly:

$$\cos y \cdot \frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} = \frac{1}{\cos y} = \frac{1}{\sqrt{1-\sin^2 y}} = \frac{1}{\sqrt{1-x^2}}$$

(Taking positive root since $-\pi/2 \le y \le \pi/2$ so $\cos y \ge 0$.)

3.2 Chain Rule Extensions

Function	Derivative
$\arcsin\left(\frac{x}{a}\right)$	$\frac{1}{\sqrt{a^2-x^2}}$
$\arctan\left(\frac{x}{a}\right)$	$\frac{a}{a^2+x^2}$
$\arcsin(f(x))$	$\frac{f'(x)}{\sqrt{1-(f(x))^2}}$

Example: Differentiate $y = \arctan(3x^2)$.

$$\frac{dy}{dx} = \frac{6x}{1+(3x^2)^2} = \frac{6x}{1+9x^4}$$

4. Integration Using Inverse Trig Functions

$$\int \frac{1}{\sqrt{a^2-x^2}}\,dx = \arcsin\left(\frac{x}{a}\right) + C, \quad |x| < a$$

$$\int \frac{1}{a^2+x^2}\,dx = \frac{1}{a}\arctan\left(\frac{x}{a}\right) + C$$

Example: Evaluate $\displaystyle\int_0^1 \frac{1}{\sqrt{4-x^2}}\,dx$.

$a = 2$. $\left[\arcsin\frac{x}{2}\right]_0^1 = \arcsin\frac{1}{2} - \arcsin 0 = \frac{\pi}{6} - 0 = \frac{\pi}{6}$

Example: Evaluate $\displaystyle\int \frac{1}{9+x^2}\,dx$.

$a = 3$. $\frac{1}{3}\arctan\left(\frac{x}{3}\right) + C$

4.1 Completing the Square

When the denominator is a quadratic $ax^2 + bx + c$, complete the square first to write it in the form $(x-h)^2 + k^2$ or $k^2 - (x-h)^2$.

Example: Find $\displaystyle\int \frac{1}{x^2+4x+13}\,dx$.

Complete the square: $x^2 + 4x + 13 = (x+2)^2 + 9$.

Substitute $u = x + 2$:

$$\int \frac{du}{u^2 + 9} = \frac{1}{3} \arctan \frac{u}{3} + C = \frac{1}{3} \arctan \frac{x+2}{3} + C$$

Example: Find $\displaystyle\int \frac{1}{\sqrt{8 - 2x - x^2}}\, dx$.

Complete the square: $8 - 2x - x^2 = 9 - (x+1)^2$.
Substitute $u = x + 1$:

$$\int \frac{du}{\sqrt{9 - u^2}} = \arcsin \frac{u}{3} + C = \arcsin \frac{x+1}{3} + C$$

Exam tip: Always check whether the numerator is a constant or has a variable part. If the numerator contains x, split as $\frac{Ax+B}{ax^2+bx+c} = \frac{A}{2a} \cdot \frac{2ax+b}{ax^2+bx+c} + \frac{\text{remainder}}{ax^2+bx+c}$. The first part integrates as a logarithm, the second as arctan.

5. Partial Fractions — Repeated and Irreducible Quadratic Factors

5.1 Repeated Linear Factors

$$\frac{f(x)}{(ax+b)^2(cx+d)} = \frac{A}{ax+b} + \frac{B}{(ax+b)^2} + \frac{C}{cx+d}$$

Example: Decompose $\dfrac{3x+1}{(x-1)^2(x+2)}$.

$$\frac{3x+1}{(x-1)^2(x+2)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+2}$$

Multiply through: $3x + 1 = A(x-1)(x+2) + B(x+2) + C(x-1)^2$

$x = 1$: $4 = 3B \Rightarrow B = \frac{4}{3}$
$x = -2$: $-5 = 9C \Rightarrow C = -\frac{5}{9}$
$x = 0$: $1 = -2A + 2B + C \Rightarrow 1 = -2A + \frac{8}{3} - \frac{5}{9} \to 2A = \frac{8}{3} - \frac{5}{9} - 1 = \frac{24-5-9}{9} = \frac{10}{9} \to A = \frac{5}{9}$
Check: $A + C = 0$ (coefficient of x^2 must be 0) ✓

5.2 Irreducible Quadratic Factors

$$\frac{f(x)}{(ax^2+bx+c)(dx+e)} = \frac{Ax+B}{ax^2+bx+c} + \frac{C}{dx+e}$$

where $ax^2 + bx + c$ does not factorise over $\mathbb{R}$.

Example: Decompose $\dfrac{5x^2+4}{(x^2+4)(x-1)}$.

$$\frac{5x^2+4}{(x^2+4)(x-1)} = \frac{Ax+B}{x^2+4} + \frac{C}{x-1}$$

Multiply through: $5x^2 + 4 = (Ax+B)(x-1) + C(x^2+4)$
$x = 1$: $9 = 5C \Rightarrow C = \frac{9}{5}$
Expand and compare coefficients:
x^2: $5 = A + C \Rightarrow A = 5 - \frac{9}{5} = \frac{16}{5}$
x^0: $4 = -B + 4C = -B + \frac{36}{5} \Rightarrow B = \frac{36}{5} - 4 = \frac{16}{5}$
Check x^1: $0 = B - A = 0$ ✓

6. Integration by Parts — Repeated Application (Tabular Method)

$$\int u\,dv = uv - \int v\,du$$

For integrals requiring repeated integration by parts (e.g., $\int x^n e^x \, dx$, $\int e^{ax} \sin(bx) \, dx$), the **tabular method** (also called the DI method) is efficient.

6.1 Tabular Method for Polynomial × Exponential

Example: Find $\displaystyle\int x^3 e^x \, dx$ using the tabular method.

Derivatives (D)	Integrals (I)	Sign
x^3	e^x	$+$
$3x^2$	e^x	$-$
$6x$	e^x	$+$
6	e^x	$-$
0	e^x	$+$

Multiply diagonally with alternating signs:

$$\int x^3 e^x \, dx = x^3 e^x - 3x^2 e^x + 6x e^x - 6e^x + C = e^x(x^3 - 3x^2 + 6x - 6) + C$$

6.2 Cyclic Integration by Parts

Example: Find $\displaystyle\int e^{2x} \cos 3x \, dx$.

Let $I = \int e^{2x} \cos 3x \, dx$.
First IBP: $u = e^{2x}$, $dv = \cos 3x \, dx$, so $v = \frac{\sin 3x}{3}$:

$$I = \frac{e^{2x} \sin 3x}{3} - \int \frac{2e^{2x} \sin 3x}{3} \, dx$$

Second IBP on remaining integral: $u = e^{2x}$, $dv = \sin 3x\,dx$, $v = -\frac{\cos 3x}{3}$:

$$\int e^{2x}\sin 3x\,dx = -\frac{e^{2x}\cos 3x}{3} + \int \frac{2e^{2x}\cos 3x}{3}\,dx = -\frac{e^{2x}\cos 3x}{3} + \frac{2I}{3}$$

Substitute back:

$$I = \frac{e^{2x}\sin 3x}{3} - \frac{2}{3}\left(-\frac{e^{2x}\cos 3x}{3} + \frac{2I}{3}\right)$$

$$I = \frac{e^{2x}\sin 3x}{3} + \frac{2e^{2x}\cos 3x}{9} - \frac{4I}{9}$$

$$\frac{13I}{9} = \frac{3e^{2x}\sin 3x + 2e^{2x}\cos 3x}{9}$$

$$I = \frac{e^{2x}(3\sin 3x + 2\cos 3x)}{13} + C$$

Exam tip: For cyclic IBP ($e^{ax}\sin bx$ type), always call the original integral I, apply IBP twice keeping the same choice of "u", then solve for I. Don't forget the constant of integration.

7. Summary of Integration Formulas

Integrand	Result
$\frac{1}{\sqrt{a^2 - x^2}}$	$\arcsin\left(\frac{x}{a}\right) + C$
$\frac{1}{a^2 + x^2}$	$\frac{1}{a}\arctan\left(\frac{x}{a}\right) + C$
$\frac{1}{\sqrt{x^2 + a^2}}$	$\operatorname{arsinh}\left(\frac{x}{a}\right) + C = \ln\left(x + \sqrt{x^2 + a^2}\right) + C$
$\frac{1}{\sqrt{x^2 - a^2}}$	$\operatorname{arcosh}\left(\frac{x}{a}\right) + C = \ln\left(x + \sqrt{x^2 - a^2}\right) + C$
$\frac{1}{a^2 - x^2}$	$\frac{1}{2a}\ln\left\lvert\frac{a+x}{a-x}\right\rvert + C$

8. Exam-Style Practice Questions

1. **[Improper integral]** Determine whether $\int_0^\infty xe^{-x}\,dx$ converges, and if so, find its value. (Hint: use integration by parts.)
2. **[Improper integral]** Show that $\int_0^1 \frac{1}{x^{2/3}}\,dx$ converges and find its value.
3. **[Mean value]** Find the mean value of $f(x) = e^{2x}$ over the interval $[0, 3]$.
4. **[Inverse trig integration]** Find $\int \frac{3}{\sqrt{16-9x^2}}\,dx$.
5. **[Completing the square]** Evaluate $\int_0^1 \frac{2x+3}{x^2+2x+5}\,dx$.
6. **[Partial fractions]** Show that $\int \frac{4x^2+1}{(x^2+1)(2x-1)}\,dx = \ln|2x-1| + \arctan x + C$.
7. **[IBP tabular]** Find $\int x^2 \sin x\,dx$.
8. **[Cyclic IBP]** Find $\int e^{-x}\sin 2x\,dx$.

General exam tips for CP2.3:

- Improper integrals: always write $\lim_{t\to\infty}$ explicitly and evaluate before taking the limit
- For inverse trig integrals, identify a correctly: in $\frac{1}{a^2+x^2}$, a is the square root of the constant
- Completing the square: write out every algebraic step — errors here lose all subsequent marks
- IBP cyclic: rearranging to get I on one side is the key step — don't omit the $+C$ at the end